

LICENSURE EXAMINATION FOR TEACHERS (LET)

Refresher Course

Focus: Professional Education
Area: Assessment of Student Learning

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PART I: Content Update

BASIC CONCEPTS

Test

- An instrument designed to measure any quality, ability, skill or knowledge.
- Comprised of test items of the area it is designed to measure.

Measurement

- A process of quantifying the degree to which someone/something possesses a given trait (i.e. quality, characteristics or features)
- A process by which traits, characteristics and behaviors are differentiated.

Assessment

- A process of gathering and organizing data into an interpretable form to have basis for decision-making
- It is a prerequisite to evaluation. It provides the information which enables evaluation to take place.

Evaluation

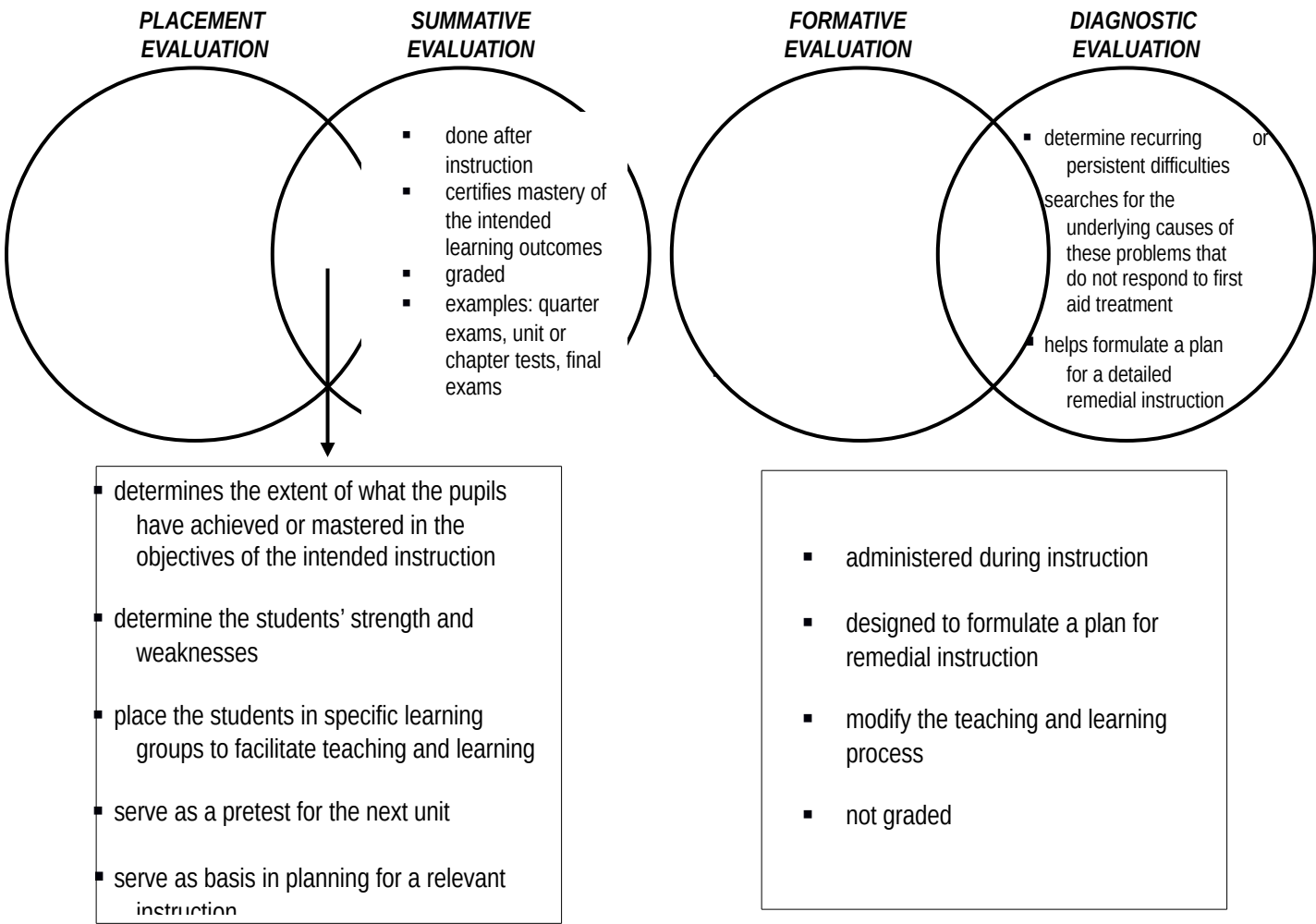
- A process of systematic analysis of both qualitative and quantitative data in order to make sound judgment or decision.
- It involves judgment about the desirability of changes in students.

MODES OF ASSESSMENT

MODE	DESCRIPTION	EXAMPLES	ADVANTAGES	DISADVANTAGES
Traditional	The objective paper-and-pen test which usually assesses low-level thinking skills	▪ Standardized Tests ▪ Teacher-made Tests	▪ Scoring is objective ▪ Administration is easy because students can take the test at the same time	▪ Preparation of instrument is time-consuming ▪ Prone to cheating
Performance	A mode of assessment that requires actual demonstration of skills or creation of products of learning	▪ Practical Test ▪ Oral and Aural Tests ▪ Projects	▪ Preparation of the instrument is relatively easy ▪ Measures behaviours that cannot be deceived	▪ Scoring tends to be subjective without rubrics ▪ Administration is time consuming
Portfolio	A process of gathering multiple indicators of student progress to support course goals in dynamic, ongoing and	▪ Working Portfolios ▪ Show Portfolios ▪ Documentary	▪ Measures student's growth and development	▪ Development is time consuming ▪ Rating tends to be subjective without

	collaborative process	Portfolios	▪ Intelligence-fair	rubrics
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FOUR TYPES OF EVALUATION PROCEDURES



PRINCIPLES OF HIGH QUALITY ASSESSMENT

- 1) **Clarity of Learning Targets**
 - Clear and appropriate learning targets include (1) what students know and can do and (2) the criteria for judging student performance.
- 2) **Appropriateness of Assessment Methods**
 - The method of assessment to be used should match the learning targets.
- 3) **Validity**
 - This refers to the degree to which a score-based inference is appropriate, reasonable, and useful.
- 4) **Reliability**
 - This refers to the degree of consistency when several items in a test measure the same thing, and stability when the same measures are given across time.
- 5) **Fairness**
 - Fair assessment is unbiased and provides students with opportunities to demonstrate what they have learned.
- 6) **Positive Consequences**
 - The overall quality of assessment is enhanced when it has a positive effect on student motivation and study habits. For the teachers, high-quality assessments lead to better information and decision-making about students.
- 7) **Practicality and efficiency**
 - Assessments should consider the teacher’s familiarity with the method, the time required, the complexity of administration, the ease of scoring and interpretation, and cost.

INSTRUCTIONAL OBJECTIVES

LEARNING TAXONOMIES

A. COGNITIVE DOMAIN

Levels of Learning Outcomes	Description	Some Question Cues
Knowledge	Involves remembering or recalling previously learned material or a wide range of materials	List, define, identify, name, recall, state, arrange
Comprehension	Ability to grasp the meaning of material by translating material from one form to another or by interpreting material	Describe, interpret, classify, differentiate, explain, translate
Application	Ability to use learned material in new and concrete situations	Apply, demonstrate, solve, interpret, use, experiment
Analysis	Ability to break down material into its component parts so that the whole structure is understood	Analyse, separate, explain, examine, discriminate, infer
Synthesis	Ability to put parts together to form a new whole	Integrate, plan, generalize, construct, design, propose
Evaluation	Ability to judge the value of material on the basis of a definite criteria	Assess, decide, judge, support, summarize, defend

B. AFFECTIVE DOMAIN

Categories	Description	Some Illustrative Verbs
Receiving	Willingness to receive or to attend to a particular phenomenon or stimulus	Acknowledge, ask, choose, follow, listen, reply, watch
Responding	Refers to active participation on the part of the student	Answer, assist, contribute, cooperate, follow-up, react
Valuing	Ability to see worth or value in a subject, activity, etc.	Adopt, commit, desire, display, explain, initiate, justify, share
Organization	Bringing together a complex of values, resolving conflicts between them, and beginning to build an internally consistent value system	Adapt, categorize, establish, generalize, integrate, organize
Value Characterization	Values have been internalized and have controlled ones' behaviour for a sufficiently long period of time	Advocate, behave, defend, encourage, influence, practice

C. PSYCHOMOTOR DOMAIN

Categories	Description	Some Illustrative Verbs
Imitation	Early stages in learning a complex skill after an indication of readiness to take a particular type of action.	Carry out, assemble, practice, follow, repeat, sketch, move
Manipulation	A particular skill or sequence is practiced continuously until it becomes habitual and done with some confidence and proficiency.	(same as imitation) - acquire, complete, conduct, improve, perform, produce
Precision	A skill has been attained with proficiency and efficiency.	(same as imitation and manipulation) - Achieve, accomplish, excel, master, succeed, surpass
Articulation	An individual can modify movement patterns to a meet a particular situation.	Adapt, change, excel, reorganize, rearrange, revise

Naturalization	An individual responds automatically and creates new motor acts or ways of manipulation out of understandings, abilities, and skills developed.	Arrange, combine, compose, construct, create, design
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DIFFERENT TYPES OF TESTS

MAIN POINTS FOR COMPARISON	TYPES OF TESTS	
Purpose	Psychological	Educational
	Aims to measure students intelligence or mental ability in a large degree without reference to what the students has learned (e.g. Aptitude Tests, Personality Tests, Intelligence Tests)	Aims to measure the result of instructions and learning (e.g. Achievement Tests, Performance Tests)

Scope of Content	Survey	Mastery
	Covers a broad range of objectives	Covers a specific objective
	Measures general achievement in certain subjects	Measures fundamental skills and abilities
	Constructed by trained professional	Typically constructed by the teacher

Language Mode	Verbal	Non-Verbal
	Words are used by students in attaching meaning to or responding to test items	Students do not use words in attaching meaning to or in responding to test items

Construction	Standardized	Informal
	Constructed by a professional item writer	Constructed by a classroom teacher
	Covers a broad range of content covered in a subject area	Covers a narrow range of content
	Uses mainly multiple choice	Various types of items are used
	Items written are screened and the best items were chosen for the final instrument	Teacher picks or writes items as needed for the test
	Can be scored by a machine	Scored manually by the teacher
	Interpretation of results is usually norm-referenced	Interpretation is usually criterion-referenced

Manner of Administration	Individual	Group
	Mostly given orally or requires actual demonstration of skill	This is a paper-and-pen test
	One-on-one situations, thus, many opportunities for clinical observation	Loss of rapport, insight and knowledge about each examinee
	Chance to follow-up examinee’s response in order to clarify or comprehend it more clearly	Same amount of time needed to gather information from one student

Effect of Biases	Objective	Subjective
	Scorer’s personal judgment does not affect the scoring	Affected by scorer’s personal opinions, biases and judgments

	▪ Worded that only one answer is acceptable	▪ Several answers are possible
	▪ Little or no disagreement on what is the correct answer	▪ Possible to disagreement on what is the correct answer
Time Limit and Level of Difficulty	Power	Speed
	▪ Consists of series of items arranged in ascending order of difficulty	▪ Consists of items approximately equal in difficulty
	▪ Measures student's ability to answer more and more difficult items	▪ Measure's student's speed or rate and accuracy in responding
Format	Selective	Supply
	▪ There are choices for the answer	▪ There are no choices for the answer
	▪ Multiple choice, True or False, Matching Type	▪ Short answer, Completion, Restricted or Extended Essay
	▪ Can be answered quickly	▪ May require a longer time to answer
	▪ Prone to guessing	▪ Less chance to guessing but prone to bluffing
	▪ Time consuming to construct	▪ Time consuming to answer and score
Nature of Assessment	Maximum Performance	Typical Performance
	▪ Determines what individuals can do when performing at their best	▪ Determines what individuals will do under natural conditions
Interpretation	Norm-Referenced	Criterion-Referenced
	▪ Result is interpreted by comparing one student's performance with other students' performance	▪ Result is interpreted by comparing student's performance based on a predefined standard (mastery)
	▪ Some will really pass	▪ All or none may pass
	▪ There is competition for a limited percentage of high scores	▪ There is no competition for a limited percentage of high score
	▪ Typically covers a <u>large</u> domain of learning tasks	▪ Typically focuses on a <u>delimited</u> domain of learning tasks
	▪ Emphasizes <u>discrimination</u> among individuals in terms of level of learning	▪ Emphasizes <u>description</u> of what learning tasks individuals can and cannot perform
	▪ Favors items of average difficulty and typically omits very easy and very hard items	▪ Matches item difficulty to learning tasks, without altering item difficulty or omitting easy or hard items
	▪ Interpretation requires a clearly defined group	▪ Interpretation requires a clearly defined and delimited achievement domain

Four Commonly-used References for Classroom Interpretation

Reference	Interpretation Provided	Condition That Must Be Present
Ability-referenced	How are students performing relative to what they are capable of doing?	Good measures of the students' maximum possible performance
Growth-referenced	How much have students changed or improved relative to what they were doing earlier?	Pre- and Post- measures of performance that are highly reliable

Norm-referenced	How well are students doing with respect to what is typical or reasonable?	Clear understanding of whom students are being compared to
Criterion-referenced	What can students do and not do?	Well-defined content domain that was assessed.

TYPES OF TEST ACCORDING TO FORMAT

1. **Selective Type** – provides choices for the answer
 - a. **Multiple Choice** – consists of a stem which describes the problem and 3 or more alternatives which give the suggested solutions. The incorrect alternatives are the distractors.
 - b. **True-False or Alternative Response** – consists of declarative statement that one has to mark true or false, right or wrong, correct or incorrect, yes or no, fact or opinion, and the like.
 - c. **Matching Type** – consists of two parallel columns: Column A, the column of premises from which a match is sought; Column B, the column of responses from which the selection is made.

Type	Advantages	Limitations
Multiple Choice	- More adequate sampling of content - Tend to structure the problem to be addressed more effectively - Can be quickly and objectively scored	- Prone to guessing - Often indirectly measure targeted behaviors - Time-consuming to construct
Alternative Response	- More adequate sampling of content - Easy to construct - Can be effectively and objectively scored	- Prone to guessing - Can be used only when dichotomous answers represent sufficient response options - Usually must indirectly measure performance related to procedural knowledge
Matching Type	- Allows comparison of related ideas, concepts, or theories - Effectively assesses association between a variety of items within a topic - Encourages integration of information - Can be quickly and objectively scored - Can be easily administered	- Difficult to produce a sufficient number of plausible premises - Not effective in testing isolated facts - May be limited to lower levels of understanding - Useful only when there is a sufficient number of related items - May be influenced by guessing

2. **Supply Test**
 - a. **Short Answer** – uses a direct question that can be answered by a word, phrase, a number, or a symbol
 - b. **Completion Test** – consists of an incomplete statement

Advantages	Limitations
- Easy to construct - Require the student to supply the answer - Many can be included in one test	- Generally limited to measuring recall of information - More likely to be scored erroneously due to a variety of responses

3. **Essay Test**
 - a. **Restricted Response** – limits the content of the response by restricting the scope of the topic
 - b. **Extended Response** – allows the students to select any factual information that they think is pertinent, to organize their answers in accordance with their best judgment

Advantages	Limitations
▪ Measure more directly behaviors specified by performance objectives ▪ Examine students' written communication skills ▪ Require the student to supply the response	▪ Provide a less adequate sampling of content ▪ Less reliable scoring ▪ Time-consuming to score

GENERAL SUGGESTIONS IN WRITING TESTS

1. Use your test specifications as guide to item writing.
2. Write more test items than needed.
3. Write the test items well in advance of the testing date.
4. Write each test item so that the task to be performed is clearly defined.
5. Write each test item in appropriate reading level.
6. Write each test item so that it does not provide help in answering other items in the test.
7. Write each test item so that the answer is one that would be agreed upon by experts.
8. Write test items so that it is the proper level of difficulty.
9. Whenever a test is revised, recheck its relevance.

SPECIFIC SUGGESTIONS

A. SUPPLY TYPE

1. Word the item/s so that the required answer is both brief and specific.
2. Do not take statements directly from textbooks to use as a basis for short answer items.
3. A direct question is generally more desirable than an incomplete statement.
4. If the item is to be expressed in numerical units, indicate type of answer wanted.
5. Blanks should be equal in length.
6. Answers should be written before the item number for easy checking.
7. When completion items are to be used, do not have too many blanks. Blanks should be at the center of the sentence and not at the beginning.

Essay Type

1. Restrict the use of essay questions to those learning outcomes that cannot be satisfactorily measured by objective items.
2. Formulate questions that will cell forth the behavior specified in the learning outcome.
3. Phrase each question so that the pupils' task is clearly indicated.
4. Indicate an approximate time limit for each question.
5. Avoid the use of optional questions.

B. SELECTIVE TYPE

Alternative-Response

1. Avoid broad statements.
2. Avoid trivial statements.
3. Avoid the use of negative statements especially double negatives.
4. Avoid long and complex sentences.
5. Avoid including two ideas in one sentence unless cause and effect relationship is being measured.
6. If opinion is used, attribute it to some source unless the ability to identify opinion is being specifically measured.
7. True statements and false statements should be approximately equal in length.
8. The number of true statements and false statements should be approximately equal.
9. Start with false statement since it is a common observation that the first statement in this type is always positive.

Matching Type

1. Use only homogenous materials in a single matching exercise.
2. Include an unequal number of responses and premises, and instruct the pupils that response may be used once, more than once, or not at all.
3. Keep the list of items to be matched brief, and place the shorter responses at the right.
4. Arrange the list of responses in logical order.
5. Indicate in the directions the bass for matching the responses and premises.
6. Place all the items for one matching exercise on the same page.

Multiple Choice

1. The stem of the item should be meaningful by itself and should present a definite problem.
2. The item should include as much of the item as possible and should be free of irrelevant information.
3. Use a negatively stated item stem only when significant learning outcome requires it.
4. Highlight negative words in the stem for emphasis.
5. All the alternatives should be grammatically consistent with the stem of the item.
6. An item should only have one correct or clearly best answer.
7. Items used to measure understanding should contain novelty, but beware of too much.
8. All distracters should be plausible.
9. Verbal association between the stem and the correct answer should be avoided.
10. The relative length of the alternatives should not provide a clue to the answer.
11. The alternatives should be arranged logically.
12. The correct answer should appear in each of the alternative positions and approximately equal number of times but in random number.
13. Use of special alternatives such as “none of the above” or “all of the above” should be done sparingly.
14. Do not use multiple choice items when other types are more appropriate.
15. Always have the stem and alternatives on the same page.
16. Break any of these rules when you have a good reason for doing so.

ALTERNATIVE ASSESSMENT

PERFORMANCE AND AUTHENTIC ASSESSMENTS

<i>When To Use</i>	▪ Specific behaviors or behavioural outcomes are to be observed ▪ Possibility of judging the appropriateness of students’ actions ▪ A process or outcome cannot be directly measured by paper-&-pencil tests
<i>Advantages</i>	▪ Allow evaluation of complex skills which are difficult to assess using written tests ▪ Positive effect on instruction and learning ▪ Can be used to evaluate both the process and the product
<i>Limitations</i>	▪ Time-consuming to administer, develop, and score ▪ Subjectivity in scoring ▪ Inconsistencies in performance on alternative skills

PORTFOLIO ASSESSMENT

- Characteristics:
1. Adaptable to individualized instructional goals
 2. Focus on assessment of products
 3. Identify students’ strengths rather than weaknesses
 4. Actively involve students in the evaluation process
 5. Communicate student achievement to others
 6. Time-consuming
 7. Need of a scoring plan to increase reliability

TYPES	DESCRIPTION
Showcase	▪ A collection of students’ best work

Reflective	▪ Used for helping teachers, students, and family members think about various dimensions of student learning (e.g. effort, achievement, etc.)
Cumulative	▪ A collection of items done for an extended period of time ▪ Analyzed to verify changes in the products and process associated with student learning
Goal-based	▪ A collection of works chosen by students and teachers to match pre-established objectives
Process	▪ A way of documenting the steps and processes a student has done to complete a piece of work

RUBRICS

→ scoring guides, consisting of specific pre-established performance criteria, used in evaluating student work on performance assessments

Two Types:

1. **Holistic Rubric** – requires the teacher to score the overall process or product as a whole, without judging the component parts separately
2. **Analytic Rubric** – requires the teacher to score individual components of the product or performance first, then sums the individual scores to obtain a total score

AFFECTIVE ASSESSMENTS

1. **Closed-Item or Forced-choice Instruments** – ask for one or specific answer
 - a. **Checklist** – measures students’ preferences, hobbies, attitudes, feelings, beliefs, interests, etc. by marking a set of possible responses
 - b. **Scales** – these instruments that indicate the extent or degree of one’s response
 - 1) Rating Scale – measures the degree or extent of one’s attitudes, feelings, and perception about ideas, objects and people by marking a point along 3- or 5- point scale
 - 2) Semantic Differential Scale – measures the degree of one’s attitudes, feelings and perceptions about ideas, objects and people by marking a point along 5- or 7- or 11- point scale of semantic adjectives
 - 3) Likert Scale – measures the degree of one’s agreement or disagreement on positive or negative statements about objects and people
 - c. **Alternate Response** – measures students preferences, hobbies, attitudes, feelings, beliefs, interests, etc. by choosing between two possible responses
 - d. **Ranking** – measures students preferences or priorities by ranking a set of responses
2. **Open-Ended Instruments** – they are open to more than one answer
 - a. **Sentence Completion** – measures students preferences over a variety of attitudes and allows students to answer by completing an unfinished statement which may vary in length
 - b. **Surveys** – measures the values held by an individual by writing one or many responses to a given question
 - c. **Essays** – allows the students to reveal and clarify their preferences, hobbies, attitudes, feelings, beliefs, and interests by writing their reactions or opinions to a given question

SUGGESTIONS IN WRITING NON-TEST OF ATTITUDINAL NATURE

1. Avoid statements that refer to the past rather than to the present.
2. Avoid statements that are factual or capable of being interpreted as factual.
3. Avoid statements that may be interpreted in more than one way.
4. Avoid statements that are irrelevant to the psychological object under consideration.
5. Avoid statements that are likely to be endorsed by almost everyone or by almost no one.
6. Select statements that are believed to cover the entire range of affective scale of interests.
7. Keep the language of the statements simple, clear and direct.
8. Statements should be short, rarely exceeding 20 words.
9. Each statement should contain only one complete thought.

10. Statements containing universals such as all, always, none and never often introduce ambiguity and should be avoided.
11. Words such as only, just, merely, and others of similar nature should be used with care and moderation in writing statements.
12. Whenever possible, statements should be in the form of simple statements rather than in the form of compound or complex sentences.
13. Avoid the use of words that may not be understood by those who are to be given the completed scale.
14. Avoid the use of double negatives.

CRITERIA TO CONSIDER IN CONSTRUCTING GOOD TESTS

VALIDITY - the degree to which a test measures what is intended to be measured. It is the usefulness of the test for a given purpose. It is the most important criteria of a good examination.

FACTORS influencing the validity of tests in general

- ✓ ***Appropriateness of test*** – it should measure the abilities, skills and information it is supposed to measure
- ✓ ***Directions*** – it should indicate how the learners should answer and record their answers
- ✓ ***Reading Vocabulary and Sentence Structure*** – it should be based on the intellectual level of maturity and background experience of the learners
- ✓ ***Difficulty of Items***- it should have items that are not too difficult and not too easy to be able to discriminate the bright from slow pupils
- ✓ ***Construction of Items*** – it should not provide clues so it will not be a test on clues nor should it be ambiguous so it will not be a test on interpretation
- ✓ ***Length of Test*** – it should just be of sufficient length so it can measure what it is supposed to measure and not that it is too short that it cannot adequately measure the performance we want to measure
- ✓ ***Arrangement of Items*** – it should have items that are arranged in ascending level of difficulty such that it starts with the easy ones so that pupils will pursue on taking the test
- ✓ ***Patterns of Answers*** – it should not allow the creation of patterns in answering the test

WAYS of Establishing Validity

- ✓ ***Face Validity*** – is done by examining the physical appearance of the test
- ✓ ***Content Validity*** – is done through a careful and critical examination of the objectives of the test so that it reflects the curricular objectives
- ✓ ***Criterion-related validity*** – is established statistically such that a set of scores revealed by a test is correlated with scores obtained in another external predictor or measure. Has two purposes:
 - ***Concurrent Validity*** – describes the present status of the individual by correlating the sets of scores obtained from two measures given concurrently
 - ***Predictive Validity*** – describes the future performance of an individual by correlating the sets of scores obtained from two measures given at a longer time interval
- ✓ ***Construct Validity*** – is established statistically by comparing psychological traits or factors that influence scores in a test, e.g. verbal, numerical, spatial, etc.
 - ***Convergent Validity*** – is established if the instrument defines another similar trait other than what it intended to measure (e.g. Critical Thinking Test may be correlated with Creative Thinking Test)
 - ***Divergent Validity*** – is established if an instrument can describe only the intended trait and not other traits (e.g. Critical Thinking Test may not be correlated with Reading Comprehension Test)

RELIABILITY – it refers to the consistency of scores obtained by the same person when retested using the same instrument or one that is parallel to it.

FACTORS affecting Reliability

- ✓ **Length of the test** – as a general rule, the longer the test, the higher the reliability. A longer test provides a more adequate sample of the behavior being measured and is less distorted by chance of factors like guessing.
- ✓ **Difficulty of the test** – ideally, achievement tests should be constructed such that the average score is 50 percent correct and the scores range from zero to near perfect. The bigger the spread of scores, the more reliable the measured difference is likely to be. A test is reliable if the coefficient of correlation is not less than 0.85.
- ✓ **Objectivity** – can be obtained by eliminating the bias, opinions or judgments of the person who checks the test.
- ✓ **Administrability** – the test should be administered with ease, clarity and uniformity so that scores obtained are comparable. Uniformity can be obtained by setting the time limit and oral instructions.
- ✓ **Scorability** – the test should be easy to score such that directions for scoring are clear, the scoring key is simple, provisions for answer sheets are made
- ✓ **Economy** – the test should be given in the cheapest way, which means that answer sheets must be provided so the test can be given from time to time
- ✓ **Adequacy** - the test should contain a wide sampling of items to determine the educational outcomes or abilities so that the resulting scores are representatives of the total performance in the areas measured

Method	Type of Reliability Measure	Procedure	Statistical Measure
Test-Retest	Measure of stability	Give a test twice to the same group with any time interval between sets from several minutes to several years	Pearson r
Equivalent Forms	Measure of equivalence	Give parallel forms of test at the same time between forms	Pearson r
Test-Retest with Equivalent Forms	Measure of stability and equivalence	Give parallel forms of test with increased time intervals between forms	Pearson r
Split Half	Measure of Internal Consistency	Give a test once. Score equivalent halves of the test (e.g. odd-and even numbered items)	Pearson r and Spearman-Brown Formula
Kuder-Richardson		Give the test once, then correlate the proportion/percentage of the students passing and not passing a given item	Kuder-Richardson Formula 20 and 21
Cronbach Coefficient Alpha		Give a test once. Then estimate reliability by using the standard deviation per item and the standard deviation of the test scores	Kuder-Richardson Formula 20

ITEM ANALYSIS

STEPS:

1. Score the test. Arrange the scores from highest to lowest.
2. Get the top 27% (upper group) and below 27% (lower group) of the examinees.
3. Count the number of examinees in the upper group (PT) and lower group (PB) who got each item correct.
4. Compute for the Difficulty Index of each item.

Df =

$N = \text{the total number of examinees}$

5. Compute for the Discrimination Index.

Ds =

$n = \text{the number of examinees in each group}$

INTERPRETATION

Difficulty Index (Df)

0.76 – 1.00 → very easy
0.25 – 0.75 → average
0.00 – 0.24 → very difficult

Discrimination Index (Ds)

0.40 – above → very good
0.30 – 0.39 → reasonably good
0.20 – 0.29 → marginal item
0.19 – below → poor item

SCORING ERRORS AND BIASES

- **Leniency error:** Faculty tends to judge better than it really is.
- **Generosity error:** Faculty tends to use high end of scale only.
- **Severity error:** Faculty tends to use low end of scale only.
- **Central tendency error:** Faculty avoids both extremes of the scale.
- **Bias:** Letting other factors influence score (e.g., handwriting, typos)
- **Halo effect:** Letting general impression of student influence rating of specific criteria (e.g., student’s prior work)
- **Contamination effect:** Judgment is influenced by irrelevant knowledge about the student or other factors that have no bearing on performance level (e.g., student appearance)
- **Similar-to-me effect:** Judging more favorably those students whom faculty see as similar to themselves (e.g., expressing similar interests or point of view)
- **First-impression effect:** Judgment is based on early opinions rather than on a complete picture (e.g., opening paragraph)
- **Contrast effect:** Judging by comparing student against other students instead of established criteria and standards
- **Rater drift:** Unintentionally redefining criteria and standards over time or across a series of scorings (e.g., getting tired and cranky and therefore more severe, getting tired and reading more quickly/leniently to get the job done)

FOUR TYPES OF MEASUREMENT SCALES

<i>Measurement</i>	<i>Characteristics</i>	<i>Examples</i>
Nominal	Groups and labal data	Gender (1-male; 2-female)
Ordinal	Rank data Distance between points are indefinite	Income (1-low, 2-average, 3-high)
Interval	Distance between points are equal No absolute zero	Test scores Temperature
Ratio	Absolute zero	Height Weight

SHAPES OF FREQUENCY POLYGONS

1. *Normal / Bell-Shaped / Symmetrical*
2. *Positively Skewed* – most scores are below the mean and there are extremely high scores
3. *Negatively Skewed* – most scores are above the mean and there are extremely low scores
4. *Leptokurtic* – highly peaked and the tails are more elevated above the baseline
5. *Mesokurtic* – moderately peaked
6. *Platykurtic* – flattened peak
7. *Bimodal Curve* – curve with 2 peaks or modes
8. *Polymodal Curve* – curve with 3 or more modes

9. Rectangular Distribution – there is no mode

DESCRIBING AND INTERPRETING TEST SCORES

MEASURES OF CENTRAL TENDENCY AND VARIABILITY

ASSUMPTIONS WHEN USED	APPROPRIATE STATISTICAL TOOLS	
	MEASURES OF CENTRAL TENDENCY <i>(describes the representative value of a set of data)</i>	MEASURES OF VARIABILITY <i>(describes the degree of spread or dispersion of a set of data)</i>
- When the frequency distribution is regular or symmetrical (normal) - Usually used when data are numeric (interval or ratio)	Mean – the arithmetic average	Standard Deviation – the root-mean-square of the deviations from the mean
- When the frequency distribution is irregular or skewed - Usually used when the data is ordinal	Median – the middle score in a group of scores that are ranked	Quartile Deviation – the average deviation of the 1 st and 3 rd quartiles from the median
- When the distribution of scores is normal and quick answer is needed - Usually used when the data are nominal	Mode – the most frequent score	Range – the difference between the highest and the lowest score in the distribution

How to Interpret the Measures of Central Tendency

- The value that represents a set of data will be the basis in determining whether the group is performing better or poorer than the other groups.

How to Interpret the Standard Deviation

- The result will help you determine if the group is homogeneous or not.
- The result will also help you determine the number of students that fall below and above the average performance.

Main points to remember:

Points above Mean + 1SD = range of **above average**
Mean + 1SD } = give the limits of an **average ability**
Mean - 1SD }
Points below Mean – 1SD = range of **below average**

How to Interpret the Quartile Deviation

- The result will help you determine if the group is homogeneous or not.
- The result will also help you determine the number of students that fall below and above the average performance.

Main points to remember:

Points above Median + 1QD = range of **above average**

Median + 1QD }
Median – 1QD } = give the limits of an **average ability**

Points below Median – 1QD = range of **below average**

MEASURES OF CORRELATION

Pearson r

$$r = \frac{\frac{\sum XY}{N} - \left(\frac{\sum X}{N}\right)\left(\frac{\sum Y}{N}\right)}{\sqrt{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2} \sqrt{\frac{\sum Y^2}{N} - \left(\frac{\sum Y}{N}\right)^2}}$$

Where:

X – scores in a test
Y – scores in a retest
N – number of examinees

Spearman Brown Formula

$$\text{reliability of the whole test} = \frac{2r_{oe}}{1+r_{oe}}$$

Where:

r_{oe} – reliability coefficient using
split-half or odd-even
procedure

Kuder-Richardson Formula 20

$$KR_{20} = \frac{K}{K-1} \left[1 - \frac{\sum pq}{S^2} \right]$$

Where:

K – number of items of a test
p – proportion of the examinees
who got the item right
q – proportion of the examinees
who got the item wrong
 S^2 – variance or standard deviation
squared

Kuder-Richardson Formula 21

$$KR_{21} = \frac{K}{K-1} \left[1 - \frac{K\bar{p}q}{S^2} \right]$$

Where:

$\bar{p} = \frac{\sum X}{K}$
 $q = 1 - p$

**INTERPRETATION OF THE Pearson r
Correlation value**

- 1 ----- Perfect Positive Correlation
high positive correlation
- 1.5 ----- Positive Correlation
low positive correlation
- 1 ----- Zero Correlation
low negative correlation
- 0.5 ----- Negative Correlation

for Validity:

computed *r* should be at least 0.75
to be significant

for Reliability:

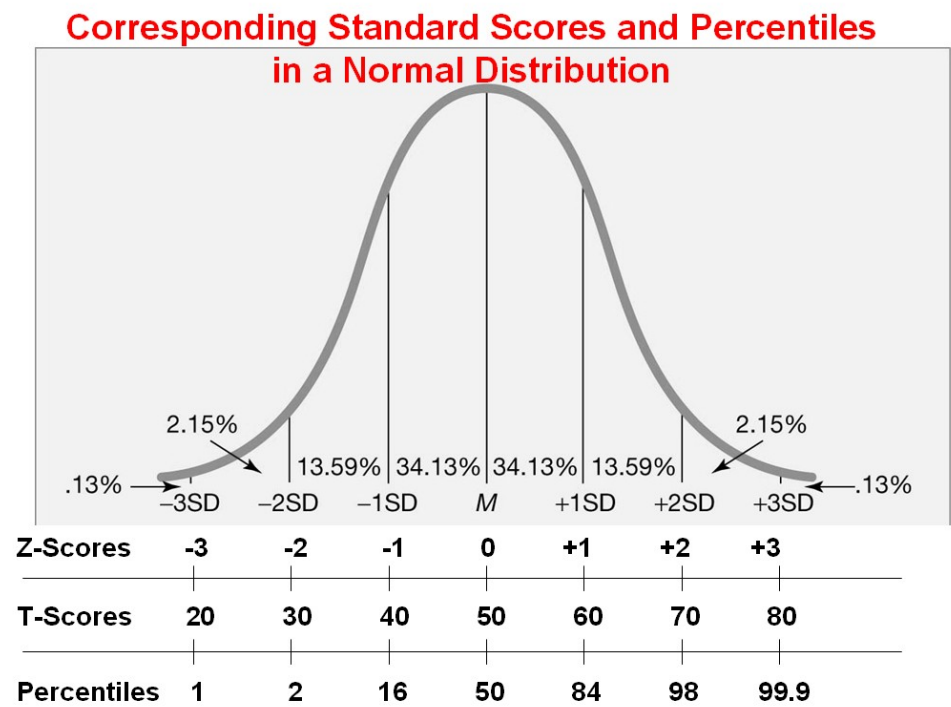
computed *r* should be at least 0.85
to be significant

high negative correlation

-1 ----- Perfect Negative Correlation

STANDARD SCORES

- Indicate the pupil’s relative position by showing how far his raw score is above or below average
- Express the pupil’s performance in terms of standard unit from the mean
- Represented by the normal probability curve or what is commonly called the normal curve
- Used to have a common unit to compare raw scores from different tests



PERCENTILE

- tells the percentage of examines that lies below one’s score

Example:

P₈₅ = 70 (This means the person who scored 70 performed better than 85% of the examinees)

Formula:

$$P_{85} = LL + i \left(\frac{85\%N - CFb}{F_{P85}} \right)$$

Z-SCORES

- tells the number of standard deviations equivalent to a given raw score

Formula:

$$Z = \frac{X - \bar{X}}{SD}$$

Where:

- X – individual’s raw score
- $\bar{X}$ – mean of the normative group
- SD – standard deviation of the normative group

Example:

Mean of a group in a test: $\bar{X} = 26$
SD = 2

Joseph's Score: $X = 27$

$$Z = \frac{X - \bar{X}}{SD} = \frac{27 - 26}{2} = \frac{1}{2}$$

$Z = 0.5$

John's Score: $X = 25$

$$Z = \frac{X - \bar{X}}{SD} = \frac{25 - 26}{2} = -\frac{1}{2}$$

$Z = -0.5$

T-SCORES

- it refers to any set of normally distributed standard deviation score that has a mean of 50 and a standard deviation of 10
- computed after converting raw scores to z-scores to get rid of negative values

Formula: **T – score = 50 + 10(Z)**

Example:

Joseph's T-score = 50 + 10(0.5)
 = 50 + 5
 = **55**

John's T-score = 50 + 10(-0.5)
 = 50 – 5
 = **45**

ASSIGNING GRADES / MARKS / RATINGS

Marking or Grading is a way to report information about a student's performance in a subject.

GRADING/REPORTING SYSTEM	ADVANTAGES	LIMITATIONS
Percentage (e.g. 70%, 86%)	▪ can be recorded and processed quickly ▪ provides a quick overview of student performance relative to other students	▪ might not actually indicate mastery of the subject equivalent to the grade ▪ too much precision
Letter (e.g. A, B, C, D, F)	▪ a convenient summary of student performance ▪ uses an optimal number of categories	▪ provides only a general indication of performance ▪ does not provide enough information for promotion
Pass – Fail	▪ encourages students to broaden their program of studies	▪ reduces the utility of grades ▪ has low reliability
Checklist	▪ more adequate in reporting student achievement	▪ time-consuming to prepare and process ▪ can be misleading at times
Written Descriptions	▪ can include whatever is relevant about the student's performance	▪ might show inconsistency between reports ▪ time-consuming to prepare and read
Parent-Teacher Conferences	▪ direct communication between parent and teacher	▪ unstructured ▪ time-consuming

GRADES:

- a. Could represent:
- how a student is performing in relation to other students (norm-referenced grading)

- the extent to which a student has mastered a particular body of knowledge (criterion-referenced grading)
 - how a student is performing in relation to a teacher's judgment of his or her potential
- b. Could be for:
- **Certification** that gives assurance that a student has mastered a specific content or achieved a certain level of accomplishment
 - **Selection** that provides basis in identifying or grouping students for certain educational paths or programs
 - **Direction** that provides information for diagnosis and planning
 - **Motivation** that emphasizes specific material or skills to be learned and helping students to understand and improve their performance
- c. Could be based on:
- | | |
|-------------------------------------|---------------------------------------|
| ▪ examination results or test data | ▪ reports, themes and research papers |
| ▪ observations of student works | ▪ discussions and debates |
| ▪ group evaluation activities | ▪ portfolios |
| ▪ class discussions and recitations | ▪ projects |
| ▪ homeworks | ▪ attitudes, etc. |
| ▪ notebooks and note taking | |
- d. Could be assigned by using:
- **Criterion-Referenced Grading** – or grading based on fixed or absolute standards where grade is assigned based on how a student has met the criteria or a well-defined objectives of a course that were spelled out in advance. It is then up to the student to earn the grade he or she wants to receive regardless of how other students in the class have performed. This is done by transmuting test scores into marks or ratings.
 - **Norm-Referenced Grading** – or grading based on relative standards where a student's grade reflects his or her level of achievement relative to the performance of other students in the class. In this system, the grade is assigned based on the average of test scores.
 - **Point or Percentage Grading System** whereby the teacher identifies points or percentages for various tests and class activities depending on their importance. The total of these points will be the bases for the grade assigned to the student.
 - **Contract Grading System** where each student agrees to work for a particular grade according to agreed-upon standards.

▪ GUIDELINES IN GRADING STUDENTS

1. Explain your grading system to the students early in the course and remind them of the grading policies regularly.
2. Base grades on a predetermined and reasonable set of standards.
3. Base your grades on as much objective evidence as possible.
4. Base grades on the student's attitude as well as achievement, especially at the elementary and high school level.
5. Base grades on the student's relative standing compared to classmates.
6. Base grades on a variety of sources.
7. As a rule, do not change grades, once computed.
8. Become familiar with the grading policy of your school and with your colleague's standards.
9. When failing a student, closely follow school procedures.
10. Record grades on report cards and cumulative records.
11. Guard against bias in grading.
12. Keep pupils informed of their standing in the class.



PART II: Test Practice

- Directions: Read and analyze each item carefully. Then, choose the best answer to each question.
- How does measurement differ from evaluation?
 - Measurement is assigning a numerical value to a given trait while evaluation is giving meaning to the numerical value of the trait.**
 - Measurement is the process of quantifying data while evaluation is the process of organizing data.
 - Measurement is a pre-requisite of assessment while evaluation is the pre-requisite of testing.
 - Measurement is gathering data while assessment is quantifying the data gathered.
 - Miss del Sol rated her students in terms of appropriate and effective use of some laboratory equipment and measurement tools and if they are able to follow the specified procedures. What mode of assessment should Miss del Sol use?

A. Portfolio Assessment	C. Traditional Assessment
B. Journal Assessment	D. Performance-Based Assessment
 - Who among the teachers below performed a formative evaluation?
 - Ms. Olivares who asked questions when the discussion was going on to know who among her students understood what she was trying to stress.
 - Mr. Borromeo who gave a short quiz after discussing thoroughly the lesson to determine the outcome of instruction.**
 - Ms. Berces who gave a ten-item test to find out the specific lessons which the students failed to understand.
 - Mrs. Corpuz who administered a readiness test to the incoming grade one pupils.
 - St. Andrews School gave a standardized achievement test instead of giving a teacher-made test to the graduating elementary pupils. Which could have been the reason why this was the kind of test given?
 - Standardized test has items of average level of difficulty while teacher-made test has varying levels of difficulty.**
 - Standardized test uses multiple-choice format while teacher-made test uses the essay test format.
 - Standardized test is used for mastery while teacher-made test is used for survey.
 - Standardized test is valid while teacher-made tests is just reliable.

- A.
5. Which test format is best to use if the purpose of the test is to relate inventors and their inventions?
A. Short-Answer
B. True-False
C. **Matching Type**
D. Multiple Choice
- A.
6. In the parlance of index of test construction, what does TOS mean?
A. Table of Specifics
B. Terms of Specifications
C. Table of Scopes
D. **Table of Specifications**
E.
7. Here is the item:
A. *"From the data presented in the table, form generalizations that are supported by the data."*
B.
C. Under what type of question does this item fall?
D. Convergent
E. Evaluative
F. Application
G. **Divergent**
H.
8. The following are synonymous to performance objectives EXCEPT:
A. Learner's objective
B. **Instructional objective**
C. Teacher's objective
D. Behavioral objective
E.
9. Which is (are) (a) norm-referenced statement?
A. Danny performed better in spelling than 60% of his classmates.
B. Danny was able to spell 90% of the words correctly.
C. Danny was able to spell 90% of the words correctly and spelled 35 words out of 50 correctly.
D. Danny spelled 35 words out of 50 correctly.
F.
10. Which guideline in test construction is NOT observed in this test item?
G. EDGAR ALLAN POE WROTE _____.
H.
A. The length of the blank suggests the answer.
B. The central problem is not packed in the stem.
C. **It is open to more than one correct answer.**
D. The blank is at the end of the question.
I.
11. Which does NOT belong to the group?
A. **Completion**
B. Matching
C. Multiple Choice
D. Alternate Response
E.
12. A test is considered reliable if
A. it is easy to score
B. it served the purpose for which it is constructed
C. **it is consistent and stable**
D. it is easy to administer
E.
13. Which is claimed to be the overall advantage of criterion-referenced over norm-referenced interpretation?
A. **An individual's score is compared with the set mastery level.**
B. An individual's score is compared with that of his peers.
C. An individual's score is compared with the average scores.
D. An individual's score does not need to be compared with any measure.
E.
14. Teacher Liza does norm-referenced interpretation of scores. Which of the following does she do?
A. She uses a specified content as its frame of reference.
B. She describes group of performance in relation to a level of master set.
C. **She compares every individual student score with others' scores.**
D. She describes what should be their performance.
E.

15. All examinees obtained scores below the mean. A graphic representation of the score distribution will be _____.
- A. negatively skewed
 - B. perfect normal curve
 - C. leptokurtic
 - D. positively skewed**
 - E.
16. In a normal distribution curve, a T-score of 70 is
- A. two SDs below the mean.
 - B. two SDs above the mean**
 - C. one SD below the mean
 - D. one SD above the mean
 - E.
17. Which type of test measures higher order thinking skills?
- A. Enumeration
 - B. Matching
 - C. Completion
 - D. Analogy**
 - E.
 - F.
- 18.
- Who is the best admired for outstanding contribution to world peace?

 - A. Kissinger
 - B. Clinton
 - C. Kennedy
 - D. Mother Teresa
- A. What is WRONG with this item?
- A. Item is overly specific.
 - B. Content is trivial.
 - C. Test item is opinion- based**
 - D. There is a cue to the right answer.
 - E.
19. The strongest disadvantage of the alternate-response type of test is
- A. the demand for critical thinking
 - B. the absence of analysis
 - C. the encouragement of rote memory**
 - D. the high possibility of guessing
 - E.
20. A class is composed of academically poor students. The distribution will most likely to be
- A. leptokurtic.
 - B. skewed to the right**
 - C. skewed to the left
 - D. symmetrical
 - E.
21. Of the following types of tests, which is the most subjective in scoring?
- A. Enumeration
 - B. Matching Type
 - C. Essay**
 - D. Multiple Choice
 - E.
22. Tom's raw score in the Filipino class is 23 which is equal to the 70th percentile. What does this imply?
- A. 70% of Tom's classmates got a score lower than 23.**
 - B. Tom's score is higher than 23% of his classmates.
 - C. 70% of Tom's classmates got a score above 23.
 - D. Tom's score is higher than 23 of his classmates.
 - A.
23. Test norms are established in order to have a basis for
- A. establishing learning objectives
 - B. identifying pupil's difficulties
 - C. planning effective instructional devices**
 - D. comparing test scores
 - E.
24. The score distribution follows a normal curve. What does this mean?
- A. Most of the scores are on the -2SD
 - B. Most of the scores are on the +2SD**
 - C. The scores coincide with the mean
 - D. Most of the scores pile up between -1SD and +1SD
 - E.
25. In her conduct of item analysis, Teacher Cristy found out that a significantly greater number from the upper group of the class got test item #5 correctly. This means that the test item
- A. has a negative discriminating power
 - B. is valid
 - C. is easy**
 - D. has a positive discriminating power
 - E.

26. Mr. Reyes tasked his students to play volleyball. What learning target is he assessing?
- A. Knowledge
 - B. Skill**
 - C. Products
 - D. Reasoning
 - E.
27. Martina obtained an NSAT percentile rank of 80. This indicates that
- A. She surpassed in performance 80% of her fellow examinees**
 - B. She got a score of 80
 - C. She surpassed in performance 20% of her fellow examinees
 - D. She answered 80 items correctly
 - F.
28. Which term refers to the collection of student's products and accomplishments for a period for evaluation purposes?
- A. Anecdotal Records
 - B. Portfolio**
 - C. Observation Report
 - D. Diary
 - E.
29. Which form of assessment is consistent with the saying "The proof of the pudding is in the eating"?
- A. Contrived
 - B. Authentic**
 - C. Traditional
 - D. Indirect
 - E.
30. Which error do teachers commit when they tend to overrate the achievement of students identified by aptitude tests as gifted because they expect achievement and giftedness to go together?
- A. Generosity error**
 - B. Central Tendency Error
 - C. Severity Error
 - D. Logical Error
 - E.
31. Under which assumption is portfolio assessment based?
- A. Portfolio assessment is dynamic assessment.**
 - B. Assessment should stress the reproduction of knowledge.
 - C. An individual learner is inadequately characterized by a test score.
 - D. An individual learner is adequately characterized by a test score.
 - E.
32. Which is a valid assessment tool if I want to find out how well my students can speak extemporaneously?
- A. Writing speeches
 - B. Written quiz on how to deliver extemporaneous speech
 - C. Performance test in extemporaneous speaking**
 - D. Display of speeches delivered
 - E.
33. Teacher J discovered that her pupils are weak in comprehension. To further determine which particular skill(s) her pupils are weak in, which test should Teacher J give?
- A. Standardized Test
 - B. Placement
 - C. Diagnostic**
 - D. Aptitude Test
 - E.
34. "Group the following items according to phylum" is a thought test item on _____.
- A. inferring
 - B. classifying**
 - C. generalizing
 - D. comparing
 - E.
35. In a multiple choice test, keeping the options brief indicates_____.
- A. Inclusion in the item irrelevant clues such as the use in the correct answer
 - B. Non-inclusion of option that mean the same**
 - C. Plausibility & attractiveness of the item
 - D. Inclusion in the item any word that must otherwise repeated in each response
 - A.
36. Which will be the most authentic assessment tool for an instructional objective on working with and relating to people?
- A. Writing articles on working and relating to people
 - B. Organizing a community project**
 - C. Home visitation
 - D. Conducting a mock election

- E.
37. While she is in the process of teaching, Teacher J finds out if her students understand what she is teaching. What is Teacher J engaged in?
- A. Criterion-referenced evaluation
B. Summative Evaluation
C. **Formative Evaluation**
D. Norm-referenced Evaluation
E.
38. With types of test in mind, which does NOT belong to the group?
- A. Restricted response essay
B. Completion
C. **Multiple choice**
D. Short Answer
E.
39. Which tests determine whether the students accept responsibility for their own behavior or pass on responsibility for their own behavior to other people?
- A. Thematic tests
B. Sentence completion tests
C. Stylistic tests
D. **Locus-of-control tests**
E.
40. When writing performance objectives, which word is NOT acceptable?
- A. Manipulate
B. Delineate
C. **Comprehend**
D. Integrate
E.
41. Here is a test item: _____ is an example of a mammal.
- A.
B. What is defective with this test item?
C. It is very elementary.
D. **The blank is at the beginning of the sentence.**
E. It is a very short question.
F. It is an insignificant test item.
G.
42. "By observing unity, coherence, emphasis and variety, write a short paragraph on taking examinations." This is an item that tests the students' skill to _____.
- A. evaluate
B. comprehend
C. **synthesize**
D. recall
E.
43. Teacher A constructed a matching type of test. In her columns of items are a combination of events, people, circumstances. Which of the following guidelines in constructing matching type of test did he violate?
- A. List options in an alphabetical order
B. Make list of items homogeneous
C. Make list of items heterogeneous
D. Provide three or more options
E.
44. Read and analyze the matching type of test given below:
- Direction: Match Column A with Column B. Write only the letter of your answer on the blank of the left column.

Column A	Column B
___ 1. Jose Rizal	A. Considered the 8 th wonder of the world
___ 2. Ferdinand Marcos	B. The national hero of the Philippines
___ 3. Corazon Aquino	C. National Heroes' Day
___ 4. Manila	D. The first woman President of the Philippines
___ 5. November 30	E. The capital of the Philippines
___ 6. Banaue Rice Terraces	F. The President of the Philippines who served several terms
- K.
- L. Question: What does the test lack?
- A. Premise
B. Option
C. Distracter
D. Response
E.
45. A number of test items in a test are said to be non-discriminating. What conclusion/s can be drawn?
- I. Teaching or learning was very good.
II. The item is so easy that anyone could get it right.
III. The item is so difficult that nobody could get it.
F.

- A. I only B. I and III C. II only D. II and III
- E.
46. Measuring the work done by a gravitational force is a learning task. At what level of cognition is it?
- A. Comprehension C. Evaluation
- B. Application** D. Analysis
- E.
47. Which improvement/s should be done in this completion test item:
- A. *An example of a mammal is _____.*
- A. The blank should be longer to accommodate all possible answers.
- B. The blank should be at the beginning of the sentence.
- C. The question should have only one acceptable answer.**
- D. The item should give more clues.
- B.
48. Here is Teacher D's lesson objective: "To trace the causes of Alzheimer's disease." Which is a valid test for this particular objective?
- A. Can an Alzheimer's disease be traced to old age? Explain.
- B. To what factors can Alzheimer's disease be traced? Explain.**
- C. What is an Alzheimer's disease?
- D. Do young people also get attacked by Alzheimer's disease? Support your answer?
- E.
49. What characteristic of a good test will pupils be assured of when a teacher constructs a table of specifications for test construction purposes?
- A. Reliability C. Construct Validity
- B. Content Validity** D. Scorability
- E.
50. Study this test item. *A test is valid when _____.*
- A. *a. it measures what is purports to measure*
- B. *b. covers a broad scope of subject matter*
- C. *c. reliability of scores*
- D. *d. easy to administer*
- E.
- F. How can you improve this test item?
- A. Make the length of the options uniform.
- B. Pack the question in the stem.
- C. Make the options parallel.**
- D. Construct the options in such a way that the grammar of the sentence remains correct.
- G.
51. In taking a test, one examinee approached the proctor for clarification on what to do. This implies a problem on which characteristic of a good test?
- A. Objectivity C. Scorability
- B. Administrability** D. Economy
- E.
52. Teacher Jane wants to determine if her students' scores in the second grading is reliable. However, she has only one set of test and her students are already on their semestral break. What test of reliability can she use?
- A. Test-retest C. Equivalent Forms
- B. Split-half** D. Test-retest with equivalent forms
- E.
53. Mrs. Cruz has only one form of test and she administered her test only once. What test of reliability can she do?
- A. Test of stability C. Test of correlation
- B. Test of equivalence **D. Test of internal consistency**
- E.
- F. Use the following table to answer items 54 – 55.

G. Class Limits	H. Frequency
I. 50 – 54	J. 9
K. 45 – 49	L. 12
M. 40 – 44	N. 16

D. To motivate students to learn and master the materials they think will be covered by the achievement test.

F.

64. The computed r for English and Math score is $-.75$. What does this mean?

- A. The higher the scores in English, the higher the scores in Math.
- B. The scores in Math and English do not have any relationship.**
- C. The higher the scores in Math, the lower the scores in English.
- D. The lower the scores in English, the lower the scores in Math.

G.

65. Which statement holds TRUE to grades?

- H. Grades are _____.
- A. exact measurements of intelligence and achievement
- B. necessarily a measure of student's intelligence
- C. intrinsic motivators for learning**
- D. are a measure of achievement

I.

66. What is the advantage of using computers in processing test results?

- A. Test results can easily be assessed.
- B. Its statistical computation is accurate
- C. Its processing takes a shorter period of time**
- D. All of the above



J.

PART III: Improving Test-Taking Skills

K.

1. Which of the following steps should be completed first in planning an achievement test?

- A. Set-up a table of specifications.
- B. Go back to the instructional objectives.**
- C. Determine the length of the test.
- D. Select the type of test items to use.
- E.

2.

_____ is an example of a leafy vegetable.

- A.
- B. Why is this test item poor?
 - I. The test item does not pose a problem to the examinee.
 - II. There is a variety of possible correct answers to this item.
 - III. The language used in the question is not precise.
 - IV. The blank is near the beginning of a sentence.
- C.

- A. I and III
- B. II and IV**
- C. I and IV
- D. I and II
- E.

3. On the first day of class after introductions, the teacher administered a Misconception/Preconception Check. She explained that she wanted to know what the class as a whole already knew about the Philippines before the Spaniards came. The Misconception/Preconception Check is a form of a

- A. diagnostic test
- B. placement test**
- C. criterion-referenced test
- D. achievement test
- F.

4. A test item has a difficulty index of $.81$ and discrimination index of $.13$. What should the test constructor do?

- A. Retain the item.
- B. Make it a bonus item.
- C. Revise the item.
- D. Reject the item.**
- G.

5. If a teacher wants to measure her students' ability to discriminate, which of these is an appropriate type of test item as implied by the direction?
 - A. "Outline the chapter on *The Cell*."
 - B. "Summarize the lesson yesterday".**
 - C. "Group the following items according to shape."
 - D. "State a set of principles that can explain the following events."
 - H.
6. A positive discrimination index means that
 - A. the test item could not discriminate between the lower and upper groups**
 - B. more from the upper group got the item correctly
 - C. more from the lower group got the item correctly
 - D. the test item has low reliability
 - I.
7. Teacher Ria discovered that her pupils are very good in dramatizing. Which tool must have helped her discover her pupil's strength?
 - A. Portfolio Assessment
 - B. Performance Assessment**
 - C. Journal Entry
 - D. Pen-and-paper Test
 - E.
8. Which among the following objectives in the psychomotor domain is highest in level?
 - A. To contract a muscle
 - B. To run a 100-meter dash
 - C. To distinguish distant and close sounds**
 - D. To dance the basic steps of the waltz
 - E.
9. If your LET items sample adequately the competencies listed in education courses syllabi, it can be said that LET possesses _____ validity.
 - A. Concurrent
 - B. Construct
 - C. Content**
 - D. Predictive
 - E.
10. In the context on the theory on multiple intelligences, what is one weakness of the pen-and-paper test?
 - A. It is not easy to administer.
 - B. It puts the non-linguistically intelligent at a disadvantage.**
 - C. It utilizes so much time.
 - D. It lacks reliability.
 - E.
11. Which test has broad sampling of topics as strength?
 - A. Objective Test**
 - B. Short Answer Test
 - C. Essay
 - D. Problem Type
 - C.
12. Quiz is to formative as periodic is to _____.
 - A. criterion-referenced
 - B. summative test**
 - C. norm-referenced
 - D. diagnostic test
 - E.
13. What does a negatively skewed score distribution imply?
 - A. The score congregate on the left side of the normal distribution curve.
 - B. The scores are widespread.
 - C. The students must be academically poor.**
 - D. The scores congregate on the right side of the normal distribution.
 - E.
14. The criterion of success in Teacher Lyn's objective is that "the pupils must be able to spell 90% of the words correctly". Ana and 19 others correctly spelled 40 words only out of 50. This means that Teacher Lyn:
 - A. attained her objective because of her effective spelling drill
 - B. attained her lesson objective
 - C. failed to attain her lesson objective as far as the twenty pupils are concerned**
 - D. did not attain her lesson objective because of the pupil's lack of attention
 - E.
15. In group norming, percentile rank of the examinee is:
 - A. dependent on his batch of examinees.**
 - B. independent on his batch of examinees.
 - C. unaffected by skewed distribution.

- D. affected by skewed distribution.
E.
16. When a significantly greater number from the lower group gets a test item correctly, this implies that the test item
- A. is very valid
B. is not very valid
C.
C. **is not highly reliable**
D. is highly reliable
17. Which applies when there are extreme scores?
- A. The median will not be a very reliable measure of central tendency.
B. The mode will be the most reliable measure of central tendency.
C. **There is no reliable measure for central tendency.**
D. The mean will not be a very reliable measure of central tendency.
E.
18. Which statement about performance-based assessment is FALSE?
- A. **They emphasize merely process.**
B. They stress on doing, not only knowing.
C. Essay tests are an example of performance-based assessments.
D. They accentuate on process as well as product.
E.
19. If the scores of your test follow a negatively skewed distribution, what should you do?
- A. Find out _____.
- A. Why your items were easy
B. **Why most of the scores are high**
C. Why most of the scores are low
D. Why some pupils scored high
E.
20. Median is to point as standard deviation is to _____.
- A. Area
B. Volume
C. **Distance**
D. Square
E.
21. Referring to assessment of learning, which statement on the normal curve is FALSE?
- A. The normal curve may not necessarily apply to homogeneous class.
B. **When all pupils achieve as expected their learning, curve may deviate from the normal curve.**
C. The normal curve is sacred. Teachers must adhere to it no matter what.
D. The normal curve may not be achieved when every pupil acquires targeted competencies.
E.
22. Aura Vivian is one-half standard deviation above the mean of his group in arithmetic and one standard deviation above in spelling. What does this imply?
- A. She excels both in arithmetic and spelling.
B. She is better in arithmetic than in spelling.
C. **She does not excel in spelling nor in arithmetic.**
D. She is better in spelling than in arithmetic.
E.
23. You give a 100-point test, three students make scores of 95, 91 and 91, respectively, while the other 22 students in the class make scores ranging from 33 to 67. The measure of central tendency which is apt to best describe for this group of 25 is
- A. the mean
B. Median
C. **An average of the median & mode**
D. the mode
E.
24. NSAT and NEAT results are interpreted against a set of mastery level. This means that NSAT and NEAT fall under
- A. **criterion-referenced test**
B. Aptitude test
C. achievement test
D. Norm-referenced test
E.

25. Which of the following is the MOST important purpose for using achievement test? To measure the_____.
- A. **Quality & quantity of previous learning** C. Educational & vocational aptitude
 B. Quality & quantity of previous teaching D. Capacity for future learning
 E.
26. What should be AVOIDED in arranging the items of the final form of the test?
- A. **Space the items so they can be read easily**
 B. Follow a definite response pattern for the correct answers to insure ease of scoring
 C. Arrange the sections such that they progress from the very simple to very complex
 D. Keep all the items and options together on the same page.
 F.
27. What is an advantage of point system of grading?
- A. **It does away with establishing clear distinctions among students.**
 B. It is precise.
 C. It is qualitative.
 D. It emphasizes learning not objectivity of scoring.
 G.
28. Which statement on test result interpretation is CORRECT?
- A. A raw score by itself is meaningful.
 B. **A student's score is a final indication of his ability.**
 C. The use of statistical technique gives meaning to pupil's scores.
 D. Test scores do not in any way reflect teacher's effectiveness.
 H.
29. Below is a list of method used to establish the reliability of the instrument. Which method is questioned for its reliability due to practice and familiarity?
- A. Split-half C. **Test-retest**
 B. Equivalent Forms D. Kuder Richardson Formula 20
 I.
30. Q3 is to 75th percentile as median is to _____.
- A. 40th percentile C. **50th percentile**
 B. 25th percentile D. 49th percentile
 J.
31. What type of test is this:
- K. *Knee is to leg as elbow is to _____.*
 L. A. M. B. N. C. Arm O. D.
 Hand Fingers Wrist
 P.
 A. **Analogy** C. Short Answer Type
 B. Rearrangement Type D. Problem Type
 Q.
32. Which statement about standard deviation is CORRECT?
- A. The lower the SD the more spread the scores are.
 B. The higher the SD the less spread the scores are.
 C. **The higher the SD the more spread the scores are.**
 D. It is a measure of central tendency.
 R.
33. Which test items do NOT affect variability of test scores?
- A. **Test items that are a bit easy.**
 B. Test items that are moderate in difficult.
 C. Test items that are a bit difficult.
 D. Test items that every examinee gets correctly.
 S.
34. Teacher B wants to diagnose in which vowel sound(s) her students have difficulty. Which tool is most appropriate?
- A. Portfolio Assessment C. **Performance Test**
 B. Journal Entry D. Paper-and-pencil Test

- T.
35. The index of difficulty of a particular test is .10. What does this mean? My students _____.
- gained mastery over the item.
 - performed very well against expectation.
 - found that the test item was either easy nor difficult.
 - find the test item difficult.
- U.
36. Study this group of test which was administered with the following results, then answer the question that follows.
- | V. | Subject | Mean | SD | Ronnel's Score |
|----|---------|------|----|----------------|
| W. | Math | 56 | 10 | 43 |
| X. | Physics | 41 | 9 | 31 |
| Y. | English | 80 | 16 | 109 |
- Z.
- AA. In which subject(s) did Ronnel perform best in relation to the group's performance?
- Physics and Math
 - English
 - Math
 - Physics
- E.
37. Which applies when the distribution is concentrated on the left side of the curve?
- Bell curve
 - Positively skewed**
 - Leptokurtic
 - Negatively Skewed
- F.
38. Standard deviation is to variability as _____ is to central tendency.
- quartile
 - mode**
 - range
 - Pearson r
- E.
39. Danny takes an IQ test thrice and each time earns a similar score. The test is said to possess _____.
- objectivity
 - reliability**
 - validity
 - scorability
- E.
40. The test item has a discrimination index of -.38 and a difficulty index of 1.0. What does this imply to test construction? Teacher must _____.
- recast the item
 - shelve the item for future use
 - reject the item
 - retain the item
- C.
41. Here is a sample TRUE-FALSE test item: *All women have a longer life-span than men.*
- What is wrong with the test item?
 - The test item is quoted verbatim from a textbook.
 - The test item contains trivial detail.**
 - A specific determiner was used in the statement.
 - The test item is vague.
- F.
42. In which competency do my students find greatest difficulty? In the item with the difficulty index of
- 1.0
 - 0.50
 - 0.90
 - 0.10
- E.
43. "Describe the reasoning errors in the following paragraph" is a sample though question on _____.
- synthesizing
 - Applying
 - analyzing
 - summarizing
- E.
44. In a one hundred-item test, what does Ryan's raw score of 70 mean?
- He surpassed 70 of his classmate in terms of score.
 - He surpassed 30 of his classmates in terms of score.
 - He got a score above the mean.
 - He got 70 items correct.**
- E.
45. Study the table on item analysis for non-attractiveness and non-plausibility of distracters based on the results of a multiple choice tryout test in math. The letter marked with an asterisk in the correct answer.

A.	B.	C.	D.	E.
A	A	B	C	D
F. U p p e r 2 7 %	G. 1	H. 4	I. 1	J. 1
K. L o w e 2 7 %	L. 6	M. 6	N. 2	O. 0

- P.
- Q. Based on the table which is the most effective distracter?
- A. Option A B. Option C **C. Option B** D. Option D
- E.
46. Here is a score distribution:
- F. 98, 93, 93, 93, 90, 88, 87, 85, 85, 85, 70, 51, 34, 34, 34, 20, 18, 15, 12, 9, 8, 6, 3, 1.
- G.
- H. Which is a characteristic of the score distribution?
- A. Bi-modal C. Skewed to the right
- B. **Tri-modal** D. No discernible pattern
- I.
47. Which measure(s) of central tendency is (are) most appropriate when the score distribution is badly skewed?
- A. Mode C. Median
- B. Mean and mode D. Mean
- J.
48. Is it wise to practice to orient our students and parents on our grading system?
- A. No, this will court a lot of complaints later.
- B. Yes, but orientation must be only for our immediate customers, the students.**
- C. Yes, so that from the very start, students and their parents know how grades are derived.
- D. No, grades and how they are derived are highly confidential.
- K.
49. With the current emphasis on self-assessment and performance assessment, which is indispensable?
- A. Numerical grading C. Transmutation Table
- B. Paper-and-Pencil Test D. Scoring Rubric
- L.
50. "In the light of the facts presented, what is most likely to happen when ...?" is a sample thought question on _____.
- A. inferring** B. generalizing C. synthesizing D. justifying
- E.
51. With grading practice in mind, what is meant by teacher's severity error?
- A.
- B. A teacher _____.
- A. tends to look down on student's answers
- B. uses tests and quizzes as punitive measures**
- C. tends to give extremely low grades
- D. gives unannounced quizzes
- C.

52. Ms. Ramos gave a test to find out how the students feel toward their subject Science. Her first item was stated as "Science is an interesting _ _ _ _ boring subject". What kind of instrument was given?
- A. Rubric
B. Likert-Scale
C. Rating Scale
D. Semantic Differential Scale
53. Which holds true to standardized tests?
- A. **They are used for comparative purposes.**
B. They are administered differently.
C. They are scored according to different standards.
D. They are used for assigning grades.
54. What is simple frequency distribution? A graphic representation of
- A. means
B. standard deviation
C. **raw scores**
D. lowest and highest scores
55. When points in scattergram are spread evenly in all directions this means that:
- A. The correlation between two variables is positive.
B. The correlation between two variables is low.
C. The correlation between two variables is high.
D. There is no correlation between two variables.
56. Which applies when skewness is 0?
- A. Mean is greater than the median.
B. Median is greater than the mean.
C. Scores have 3 modes.
D. Scores are normally distributed.
57. Which process enhances the comparability of grades?
- A. Determining the level of difficulty of the test**
B. Constructing departmentalized examinations for each subject area
C. Using table of specifications
D. Giving more high-level questions
58. In a grade distribution, what does the normal curve mean?
- A. All students having average grades.
B. A large number of students with high grades and very few low grades.
C. A large number of more or less average students and very few students receiving low and high grades
D. A large number of students receiving low grades and very few students with high grades
59. For professional growth, which is a source of teacher performance?
- A. Self-evaluation
B. Supervisory evaluation
C. Student's evaluation
D. Peer evaluation

E.

60. The following are trends in marking and reporting system, EXCEPT:

- A. indicating strong points as well as those needing improvement
- B. conducting parent-teacher conferences as often as needed
- C. raising the passing grade from 75 to 80
- D. supplementing subject grade with checklist on traits